

Discrete Structures

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UNIT-I: Fundamental of Logic and Proof Techniques

Introduction to Discrete Structures

Discrete structures is the study of mathematical objects that are **fundamentally countable and separate**, as opposed to continuous mathematics like calculus. It forms the theoretical backbone of computer science by covering logic, proofs, sets, relations, graphs, and combinatorics. The word "discrete" means distinct and disconnected, like integers (1, 2, 3...) rather than a flowing real number line.

***Discrete Structures** (also called Discrete Mathematics) is the branch of mathematics that deals with discrete objects (countable, separate values) and the formal relationships and operations defined on them.*

Logic

It is the formal study of reasoning. In computer science, we use logic to design circuits, write conditional statements (if/else), and prove that algorithms work correctly.

***Logic** (from the Greek logos — word, reason, principle) is the study of the principles of correct reasoning. In the context of discrete mathematics and computer science, logic is a formal, mathematical system for representing and manipulating statements to determine their truth values and the validity of arguments.*

Think of it as the "algebra of truth." Just as algebra lets you manipulate numbers with rules (e.g., $x+y=y+x$), logic lets you manipulate statements with rules (e.g., "If 'it is raining' is true, then 'it is not raining' is false").

Propositions and Logical Connectives

Examples:

- "5 is greater than 3" → True proposition
- "It is raining" → Proposition (truth depends on context)
- "What time is it?" → NOT a proposition (it's a question)
- " $x + 1 = 5$ " → NOT a proposition yet (x is undefined)

A **Proposition** is a declarative sentence that is strictly **True (T)** or **False (F)**, but never both.

*A **proposition** (also called a statement, sentence, or declarative sentence) is a declarative sentence that has exactly **one** of two possible truth values: **true (T)** or **false (F)**. It cannot be both, and it cannot be neither.*

In simpler terms: A proposition is a sentence about which it makes sense to ask: "Is this true or false?" and there must be a definite answer.

The three essential properties of a proposition:

- 1. Declarative form:** It must state something (not ask, command, or exclaim).
- 2. Definite truth value:** It must be either true or false (not "sometimes," "maybe," or opinion-based without evidence).
- 3. No ambiguity:** The meaning must be clear enough to determine truth value (though we may not know it yet).

Logical Connectives combine simple propositions into complex ones:

***Logical connectives** (also called logical operators or propositional connectives) are symbols or words used to combine one or more simple propositions (statements) to form a **compound proposition**. They define the logical relationship between propositions.*

Think of them as the "verbs" or "operations" of logic — just as $+$, $-$, $\times$, $\div$ operate on numbers, logical connectives operate on truth values (True/False).

Key distinction:

- **Unary connective:** Operates on a **single** proposition (e.g., NOT).
- **Binary connective:** Operates on **two** propositions (e.g., AND, OR, IF-THEN, IFF).

Name	Symbol	Read as	True when	Example
Negation	$\neg p$	NOT p	p is false	$\neg(\text{It rains}) = \text{"It does not rain"}$
Conjunction	$p \wedge q$	p AND q	Both p and q are true	Sunny $\wedge$ Warm
Disjunction	$p \vee q$	p OR q	At least one of p, q is true	Tea $\vee$ Coffee
Exclusive OR	$p \oplus q$	p XOR q	Exactly one of p, q is true	Win $\oplus$ Lose
Implication	$p \rightarrow q$	If p then q	p is false, OR q is true	Rain $\rightarrow$ Wet ground
Biconditional	$p \leftrightarrow q$	p if and only if q	p and q have the same truth value	Even $\leftrightarrow$ Divisible by 2

Negation ($\neg$, NOT)

Definition: Negation takes a single proposition p and produces its opposite truth value. If p is true, $\neg p$ is false. If p is false, $\neg p$ is true.

Natural language equivalents:

- "It is not the case that..."
- "Not p"
- "It is false that..."
- "p is false"

Truth table:

p	$\neg p$
T	F
F	T

Programming analog:

if (!isRaining) {

 // Executes when isRaining is false

}

Examples:

- Let p : "The computer is on."
- $\neg p$: "The computer is **not** on."
- Let q : "5 is greater than 10."
- $\neg q$: "It is **not** the case that 5 is greater than 10." (which is true, because $5 > 10$ is false)

Important properties:

- **Double negation:** $\neg(\neg p) \equiv p$ (two NOTs cancel out).
- In programming, `!!x` is sometimes used to convert a value to a true boolean.

Conjunction ($\wedge$, AND)

Definition: Conjunction combines two propositions p and q and is true **only when both p and q are true**. In all other cases, it is false.

Natural language equivalents:

- "p and q"
- "p but q"
- "p moreover q"
- "p also q"
- "Both p and q"

Truth table:

p	q	$p \wedge q$
---	---	--------------

T	T	T
T	F	F
F	T	F
F	F	F

Programming analog:

```
if (user.isLoggedIn && user.hasPermission) {
    // Executes ONLY when BOTH conditions are true
}
```

Examples:

- p : "It is raining." q : "It is cold."
- $p \wedge q$: "It is raining **and** it is cold." (True only if both happen)
- p : "The file exists." q : "The file is readable."
- $p \wedge q$: "The file exists **and** is readable."

Disjunction ($\vee$, OR) — Inclusive OR

Definition: Disjunction combines two propositions p and q and is true if **at least one** of p or q is true. It is false **only when both are false**.

Natural language equivalents:

- "p or q" (inclusive sense)
- "p or q or both"

Truth table:

p	q	$p \vee q$
-----	-----	------------

T	T	T
T	F	T
F	T	T
F	F	F

Programming analog:

```
if (user.isAdminRole || user.isSuperUser) {
    // Executes if AT LEAST ONE condition is true
}
```

Examples:

- p : "The light switch is on." q : "The bulb is working."
- $p \vee q$: "The light switch is on **or** the bulb is working." (If either is true, you might expect light — but not both needed.)
- p : "I will pay by cash." q : "I will pay by card."
- $p \vee q$: "I will pay by cash **or** card (or both)."

Exclusive OR ($\oplus$, XOR)

Definition: Exclusive OR combines two propositions p and q and is true if **exactly one** of p or q is true. It is false when both are true or both are false.

Natural language equivalents:

- "Either p or q , but not both"
- " p xor q "
- " p or q exclusively"

Truth table:

p	q	$p \oplus q$
-----	-----	--------------

T	T	F
T	F	T
F	T	T
F	F	F

Programming analog:

// Many languages don't have a built-in XOR for booleans, but it can be written as:

if ((a || b) && !(a && b)) { ... } // Equivalent to a XOR b

// Or using != for booleans in some languages:

if (a != b) { ... }

Examples:

- p : "The door is open." q : "The door is closed."
- $p \oplus q$: "The door is either open **or** closed, but not both." (True in normal circumstances ;a door cannot be both open and closed simultaneously.)
- p : "I will order pizza." q : "I will order pasta."
- $p \oplus q$: "I will order **either** pizza **or** pasta, but not both."

Implication ($\rightarrow$, IF-THEN)

Definition: Implication combines two propositions p (the **antecedent** or **hypothesis**) and q (the **consequent** or **conclusion**) and is **false only when p is true and q is false**. In all other cases, it is true.

Natural language equivalents:

- "If p , then q "
- " p implies q "
- " q if p "
- " p only if q "
- " p is sufficient for q "
- " q is necessary for p "

Truth table:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Programming analog:

```

if (!p || q) {
    // This is equivalent to  $p \rightarrow q$ 
}

```

Bi-conditional ($\leftrightarrow$, IFF)

Definition: Bi-conditional combines two propositions p and q and is true when p and q have the **same truth value** (both true or both false). It is false when they differ.

Natural language equivalents:

- "p if and only if q"
- "p iff q"
- "p is equivalent to q"
- "p exactly when q"

Truth table:

p	q	$p \leftrightarrow q$
T	T	T

T	F	F
F	T	F
F	F	T

Programming analog:

```
if (a == b) {
    // Executes when a and b have the same boolean value
}
```

Translating English to Logical Expressions

Converting natural language to logic removes ambiguity.

- **Sentence:** "You can access the internet from campus only if you are a computer science major or you are not a freshman."
- **Variables:**
 - a: You can access the internet from campus.
 - c: You are a computer science major.
 - f: You are a freshman.
- **Logical Expression:**

4. English Sentences to Logical Expressions	
The key skill is identifying connectives in natural language:	
English phrasing	Logical form
"p but q"	$p \wedge q$
"Neither p nor q"	$\neg p \wedge \neg q$
"p unless q"	$\neg q \rightarrow p$
"p only if q"	$p \rightarrow q$
"p is necessary for q"	$q \rightarrow p$
"p is sufficient for q"	$p \rightarrow q$

Example: "You can access the system only if you have a password and you are an admin." $\rightarrow$ Let A = access, P = has password, D = is admin $\rightarrow A \rightarrow (P \wedge D)$

Propositional Equivalences and Laws

Two compound propositions are said to be **logically equivalent** if they have the **exact same truth value** for every possible combination of truth values of their component simple propositions.

Notation: $p \equiv q$ or $p \Leftrightarrow q$ (but note: this is a statement about logic, not a logical connective within logic — though it is closely related to the bi-conditional).

Formal definition: $p \equiv q$ if and only if the bi-conditional $p \leftrightarrow q$ is a **tautology** (true for all truth assignments).

In simple terms: Two expressions are equivalent if their truth tables are **identical** ;column by column, row by row

Why Are Propositional Equivalences Important?

Area	Why It Matters
Simplifying Code	Replace complex if conditions with simpler, equivalent ones.
Digital Circuit Design	Minimize the number of logic gates, reducing cost and power consumption.
Logical Proofs	Transform a statement into a more convenient form without changing its truth value.
Database Queries	Optimize WHERE clauses by rewriting equivalent conditions.
Compiler Optimization	Compilers use logical equivalences to simplify boolean expressions in source code.
Artificial Intelligence	Knowledge base simplification and inference.

Example: The expression $(p \wedge q) \vee (p \wedge \neg q)$ might look complicated, but it is equivalent to simply p . Recognizing this can simplify a program condition from:

The Fundamental Laws of Logical Equivalence (The "Algebra of Logic")

These are the rules you must memorize. They are to logic what the commutative, associative, and distributive laws are to arithmetic.

1. Identity Laws:

Form	Explanation
------	-------------

$p \wedge \text{True} \equiv p$	AND with True does nothing.
$p \vee \text{False} \equiv p$	OR with False does nothing.

Intuition:-

- "p AND True" = always whatever p is. (True AND True = True; False AND True = False)
- "p OR False" = always whatever p is. (True OR False = True; False OR False = False)

2. Domination Laws

Form	Explanation
$p \wedge \text{False} \equiv \text{False}$	If you AND with False, the result is always False.
$p \vee \text{True} \equiv \text{True}$	If you OR with True, the result is always True.

Intuition:

- If one input to AND is False, output is False (dominated by False).
- If one input to OR is True, output is True (dominated by True).

3. Idempotent Laws

Form	Explanation
$p \wedge p \equiv p$	ANDing a proposition with itself gives the same proposition.
$p \vee p \equiv p$	ORing a proposition with itself gives the same proposition.

Intuition: "It is raining AND it is raining" means exactly the same as "It is raining."

4. Double Negation Law

Form	Explanation
$\neg(\neg p) \equiv p$	Two NOTs cancel each other.

Intuition: "It is not true that it is not raining" means "It is raining."

5. Commutative Laws

Form	Explanation
$p \wedge q \equiv q \wedge p$	Order of AND doesn't matter.
$p \vee q \equiv q \vee p$	Order of OR doesn't matter.

Intuition: "p and q" means the same as "q and p".

6. Associative Laws

Form	Explanation
$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Grouping of AND doesn't matter.
$(p \vee q) \vee r \equiv p \vee (q \vee r)$	Grouping of OR doesn't matter.

Intuition: Parentheses can be rearranged freely for chains of the same connective.

7. Distributive Laws

Form	Explanation
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	AND distributes over OR.
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	OR distributes over AND.

Intuition: This is exactly like multiplication distributing over addition in arithmetic: $a \times (b + c) = (a \times b) + (a \times c)$. but note that OR also distributes over AND (which has no direct arithmetic analog).

8. De Morgan's Laws

Form	Explanation
$\neg(p \wedge q) \equiv \neg p \vee \neg q$	The negation of AND is OR of negations.
$\neg(p \vee q) \equiv \neg p \wedge \neg q$	The negation of OR is AND of negations.

Intuition:

- "It is **not** true that both p and q are true" means "p is false **or** q is false (or both)."
- "It is **not** true that either p or q is true" means "p is false **and** q is false."

9. Absorption Laws

Form	Explanation
$p \wedge (p \vee q) \equiv p$	AND with (p OR q) absorbs the q.
$p \vee (p \wedge q) \equiv p$	OR with (p AND q) absorbs the q.

Intuition:

- "p AND (p OR q)" = p (because if p is true, both sides true; if p false, both false regardless of q).
- "p OR (p AND q)" = p (if p is true, true; if p false, the AND is false, so both false).

10. Negation (Inverse) Laws

Form	Explanation
$p \wedge \neg p \equiv \text{False}$	A proposition and its negation cannot both be true. (Contradiction)
$p \vee \neg p \equiv \text{True}$	A proposition or its negation must be true. (Tautology — Law of Excluded Middle)

11. Implication Laws

Form	Explanation
$p \rightarrow q \equiv \neg p \vee q$	If-then can be rewritten as OR of negation of antecedent with consequent.
$p \rightarrow q \equiv \neg q \rightarrow \neg p$	Contrapositive.

12. Bi-conditional (Equivalence) Laws

Form

Explanation

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ IFF means implication both ways.

$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$ IFF means both true or both false.

Predicates, Variables, and Quantifiers

A **predicate** is a sentence that contains one or more **variables** and becomes a **proposition** (true or false) only when the variables are assigned specific values or are quantified.

In other words, a predicate is a **property** of an object or a **relationship** between objects.

Notation: $P(x), Q(x,y), R(x,y,z)$, etc.

Breaking down the word:

• **Pre-** (before) + **-dicare** (to declare) → something declared about something else.

Example: $P(x) = \text{"x is greater than 5"}$

- $P(8) \rightarrow \text{True}$
- $P(3) \rightarrow \text{False}$

Predicate	Meaning	When is it a proposition?	Truth (for given assignment)
$P(x)$: "x is a prime number."	Property of x	When x is assigned a value	$P(5) = \text{True}$, $P(4) = \text{False}$
$Q(x,y)$: "x is greater than y."	Relation between x and y	When x and y are assigned	$Q(10,3) = \text{True}$, $Q(2,5) = \text{False}$

Predicate	Meaning	When is it a proposition?	Truth (for given assignment)
$R(x)$: "x is a computer science student."	Property of x	When x is assigned	$R(\text{Alice}) = \text{True or False}$
$S(x,y,z)$: "x gave y to z."	Three-way relation	When x, y, z are assigned	$S(\text{John}, \text{book}, \text{Mary}) = \text{True/False}$

Variables:

Bound Variable

A variable x is **bound** in a logical expression if it occurs within the scope of a quantifier ($\forall x$ or $\exists x$). The quantifier "binds" the variable.

Example: In $\forall x(P(x) \rightarrow Q(x))$, both occurrences of x are bound by $\forall x$.

Free Variable

A variable x is **free** in a logical expression if there is **no** quantifier that binds it. It appears outside the scope of any $\forall x$ or $\exists x$.

Example: In $P(x) \rightarrow \forall y Q(y, x)$, the variable x is **free** (it appears but is not quantified), while y is **bound** (by $\forall y$).

Identify Free vs. Bound Variables: Step-by-Step Method

Step-by-step algorithm:

1. Identify all quantifiers in the expression ($\forall$ or $\exists$). Each has a quantified variable (e.g., $\forall x$, $\exists y$).

2. Determine the scope of each quantifier. Look at the parentheses — the scope is the subexpression immediately following the quantifier (usually the next parenthesized formula).

3. Mark bound occurrences: Any occurrence of a variable that falls within the scope of a quantifier that quantifies that same variable is **bound**.

4. Mark free occurrences: Any occurrence of a variable that is **not** within the scope of **any** quantifier that quantifies that variable is **free**.

5. A variable can be bound and free in the same expression if it appears in different places, some inside a quantifier's scope and some outside. This is allowed but often confusing; avoid it.

Worked Example 1:

$$\forall x(P(x) \wedge Q(y)) \rightarrow R(x)$$

Variable	Occurrence	Analysis	Status
x	In $P(x)$	Inside scope of $\forall x$	Bound
x	In $R(x)$	Outside scope of $\forall x$ (after the parentheses close)	Free
y	In $Q(y)$	Inside scope of $\forall x$? No ; $\forall x$ binds x , not y ; no quantifier for y	Free

Result: x appears both bound (first occurrence) and free (second occurrence). y is free.

Free vs Bound variables:

- A variable is **bound** when a quantifier is applied to it ; it no longer represents an arbitrary value.
- A variable is **free** when no quantifier governs it; the expression is not yet a proposition.

Example: In $\forall x P(x, y)$, x is **bound** (governed by $\forall$), y is **free** (open/unresolved).

Quantifiers:

A **quantifier** tells us **how many** elements in the domain satisfy a predicate. There are two main quantifiers in standard logic.

1. Universal Quantifier ($\forall$)

Symbol	Name	Meaning	English equivalents
$\forall$	Universal quantifier	"for all," "for every," "all," "each," "every"	"All dogs bark." "Every student passed." "Each computer has a processor."

Form: $\forall x P(x)$

Meaning: $P(x)$ is true for **every** element x in the domain.

Truth condition: $\forall x P(x)$ is **true** if and only if $P(x)$ is true for **every single** element in the domain. If there is **at least one** counterexample (an element for which $P(x)$ is false), the universal statement is **false**.

Example:

- Domain = all positive integers.
- $P(x)$: "x is greater than 0."
- $\forall x P(x)$ = "Every positive integer is greater than 0." → **True**.

Example with counterexample:

- Domain = all integers.
- $P(x)$: " x^2 is positive."
- $\forall x P(x)$ = "For every integer, its square is positive." → **False** because $0^2=0$, which is not positive.

2. Existential Quantifier ($\exists$)

Symbol	Name	Meaning	English equivalents
$\exists$	Existential quantifier	"there exists," "for some," "at least one," "some"	"Some students are late." "There is a solution." "At least one number is prime."

Form: $\exists x P(x)$

Meaning: There exists **at least one** element x in the domain for which $P(x)$ is true.

Truth condition: $\exists x P(x)$ is **true** if and only if there is **at least one** element in the domain that makes $P(x)$ true. If **no** element satisfies $P(x)$, the existential statement is **false**.

Example:

- Domain = all integers.
- $P(x)$: "x is even and prime."
- $\exists x P(x)$ = "There exists an integer that is both even and prime." → **True** ($x = 2$).

Example with no witness:

- Domain = all positive integers.
- $P(x)$: "x is greater than itself."
- $\exists x P(x)$ = "There exists a positive integer greater than itself." → **False** (no such number).

Logical Equivalences Involving Quantifiers

These are essential for manipulating quantified statements. They allow you to push negations through quantifiers, analogous to De Morgan's laws.

De Morgan's Laws for Quantifiers

These are the quantifier analogs of De Morgan's laws for propositional logic. They tell us how to push a negation through a quantifier.

Law	Propositional Analogy
-----	-----------------------

$\neg \forall x P(x) \equiv \exists x \neg P(x)$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$
--	--

$\neg \exists x P(x) \equiv \forall x \neg P(x)$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$
--	--

Intuition: A universal quantifier ($\forall$) acts like a **giant AND** over all elements of the domain. An existential quantifier ($\exists$) acts like a **giant OR**. De Morgan's laws apply to these large conjunctions and disjunctions.

Law 1: $\neg \forall x P(x) \equiv \exists x \neg P(x)$

In Words

Example

"It is **not** true that **all** x satisfy P ."

"Not every student passed the exam."

is equivalent to:

is equivalent to:

"**There exists** an x such that P is **false**."

"Some student **did not pass** the exam."

Proof by intuition: If the domain has elements $\{x_1, x_2, x_3, \dots\}$, then:

- $\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \dots$
- $\neg \forall x P(x) \equiv \neg(P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \dots)$
- By De Morgan: $\equiv \neg P(x_1) \vee \neg P(x_2) \vee \neg P(x_3) \vee \dots$
- That is exactly $\exists x \neg P(x)$.

Law 2: $\neg \exists x P(x) \equiv \forall x \neg P(x)$

In Words

"It is **not** true that **there exists** an x satisfying P ."

is equivalent to:

"**Every** x **fails** to satisfy P ."

Example

"No student passed the exam."

is equivalent to:

"Every student **did not pass** the exam."

Proof by intuition:

- $\exists x P(x) \equiv P(x1) \vee P(x2) \vee P(x3) \vee \dots$
- $\neg \exists x P(x) \equiv \neg (P(x1) \vee P(x2) \vee P(x3) \vee \dots)$
- By De Morgan: $\equiv \neg P(x1) \wedge \neg P(x2) \wedge \neg P(x3) \wedge \dots$
- That is exactly $\forall x \neg P(x)$.

Rules of Inference for Propositions and Quantified Statements

What Are Rules of Inference?

A **rule of inference** is a valid logical form that allows us to deduce a conclusion from a set of premises. If the premises are true, applying a valid rule of inference guarantees that the conclusion is also true.

Notation: Conclusion Premises or Premises $\therefore$ Conclusion

Example (Modus Ponens):

$\therefore qp \rightarrow q, p$

"If it rains, the ground gets wet. It is raining. Therefore, the ground is wet."

Key distinction:

- **Logical equivalence** ($\equiv$) gives us two-way rewriting rules.
- **Rule of inference** ($\therefore$) gives us one-way deduction rules (though many are reversible in practice).

Rules of Inference for Propositions

These rules apply to propositional logic (statements without quantifiers).

1. Modus Ponens (MP) ; "Affirming the antecedent"

Form	Example
$p \rightarrow q$	"If it is raining, then the ground is wet."
p	"It is raining."
$\therefore q$	"Therefore, the ground is wet."

Explanation: If we know that p implies q , and we know that p is true, then q must be true.

2. Modus Tollens (MT) ; "Denying the consequent"

Form	Example
$p \rightarrow q$	"If it is raining, then the ground is wet."
$\neg q$	"The ground is not wet."
$\therefore \neg p$	"Therefore, it is not raining."

Explanation: If p implies q , but q is false, then p cannot be true (otherwise q would be true).

3. Hypothetical Syllogism (HS) ; Chain rule

Form	Example
$p \rightarrow q$	"If it rains, the ground gets wet."
$q \rightarrow r$	"If the ground gets wet, the grass grows."
$\therefore p \rightarrow r$	"Therefore, if it rains, the grass grows."

Explanation: Implication is transitive.

4. Disjunctive Syllogism (DS)

Form	Example
$p \vee q$	"Either the switch is on or the light is off."
$\neg p$	"The switch is not on."
$\therefore q$	"Therefore, the light is off."

Alternative form	Example
$p \vee q$	"Either you pay or you leave."
$\neg q$	"You do not leave."
$\therefore p$	"Therefore, you pay."

Explanation: If at least one of p or q is true, and we know one is false, the other must be true.

5. Addition (Add)

Form	Example
p	"It is raining."
$\therefore p \vee q$	"Therefore, it is raining or snowing."

Explanation: If p is true, then " p or anything" is also true. This rule introduces a disjunction.

6. Simplification (Simp)

Form	Example
$p \wedge q$	"It is raining and cold."

Form	Example
$\therefore p$	"Therefore, it is raining."

Alternative Example

$p \wedge q$	"It is raining and cold."
$\therefore q$	"Therefore, it is cold."

Explanation: If both p and q are true, then each one individually is true.

7. Conjunction (Conj)

Form	Example
p	"It is raining."
q	"It is cold."
$\therefore p \wedge q$	"Therefore, it is raining and cold."

Explanation: If two statements are true individually, their conjunction is true.

8. Resolution (Res)

Form	Example
$p \vee q$	"Either you study or you fail."
$\neg p \vee r$	"Either you do not study or you pass."
$\therefore q \vee r$	"Therefore, either you fail or you pass."

Explanation: This is the foundation of the resolution proof method used in automated theorem proving and logic programming (Prolog).

9. Constructive Dilemma (CD)

Form	Example
$(p \rightarrow q) \wedge (r \rightarrow s)$	"If you study, you pass; if you cheat, you are caught."
$p \vee r$	"Either you study or you cheat."
$\therefore q \vee s$	"Therefore, either you pass or you are caught."

10. Destructive Dilemma (DD)

Form	Example
$(p \rightarrow q) \wedge (r \rightarrow s)$	"If it rains, ground gets wet; if it's windy, trees sway."
$\neg q \vee \neg s$	"Either the ground is not wet or the trees do not sway."
$\therefore \neg p \vee \neg r$	"Therefore, either it did not rain or it is not windy."

Rules of Inference for Quantified Statements

When we move to predicate logic, we need rules to instantiate (remove) and generalize (add) quantifiers.

1. Universal Instantiation (UI)

Form	Explanation
$\forall x P(x)$	"All humans are mortal."
$\therefore P(c)$ for any specific c	"Therefore, Socrates (a specific human) is mortal."

Meaning: If a property holds for **all** members of the domain, then it holds for any **particular** member we choose.

Restriction: c can be **any** element in the domain. No special assumptions about c .

2. Universal Generalization (UG)

Form

Explanation

$P(c)$ for an **arbitrary** c "Let c be an arbitrary integer. Suppose c is even..."

$\therefore \forall x P(x)$ "Therefore, **all** integers have that property."

Meaning: If we can prove that a property holds for an **arbitrary, representative** element c (with no special assumptions), then it holds for **all** elements.

Crucial restriction: c must be **arbitrary**. You cannot use UG if c was introduced by Existential Instantiation (see below) or if you made any special assumptions about c .

3. Existential Instantiation (EI)

Form

Explanation

$\exists x P(x)$ "There exists a student who is late."

$\therefore P(c)$ for some **new** constant c "Let's call that student c . c is late."

Meaning: If we know that **some** element satisfies P , we can give it a **temporary name** and reason about it.

Crucial restrictions:

- c must be a **new** constant that has **not** appeared earlier in the proof.
- You cannot assume anything else about c except that $P(c)$ holds.
- After using EI, you cannot use Universal Generalization on c (because c is not arbitrary — it's the specific one that exists).

4. Existential Generalization (EG)

Form

Explanation

$P(c)$ for some specific c "Socrates is mortal."

$\therefore \exists x P(x)$ "Therefore, **someone** is mortal."

Meaning: If we have a specific example that satisfies P , then we can conclude that there exists at least one element satisfying P .

Quick Reference Card

Rule	Form
Modus Ponens (MP)	$p \rightarrow q, p \therefore q$
Modus Tollens (MT)	$p \rightarrow q, \neg q \therefore \neg p$
Hypothetical Syllogism (HS)	$p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$
Disjunctive Syllogism (DS)	$p \vee q, \neg p \therefore q$
Addition (Add)	$p \therefore p \vee q$
Simplification (Simp)	$p \wedge q \therefore p$
Conjunction (Conj)	$p, q \therefore p \wedge q$
Resolution (Res)	$p \vee q, \neg p \vee r \therefore q \vee r$
Universal Instantiation (UI)	$\forall x P(x) \therefore P(c)$
Universal Generalization (UG)	$P(c) \text{ (arbitrary)} \therefore \forall x P(x)$
Existential Instantiation (EI)	$\exists x P(x) \therefore P(c) \text{ (new } c)$
Existential Generalization (EG)	$P(c) \therefore \exists x P(x)$

Rule	Form	Example
PROPOSITIONAL RULES		
Modus Ponens	$p, p \rightarrow q \therefore q$	If it rains, ground is wet. It rains. $\therefore$ Ground is wet.
Modus Tollens	$\neg q, p \rightarrow q \therefore \neg p$	Ground is not wet. If it rains, ground gets wet. $\therefore$ It didn't rain.
Hypothetical Syllogism	$p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$	If A then B. If B then C. $\therefore$ If A then C.
Disjunctive Syllogism	$p \vee q, \neg p \therefore q$	Either tea or coffee. Not tea. $\therefore$ Coffee.
Addition	$p \therefore p \vee q$	It is sunny. $\therefore$ It is sunny or windy.
Simplification	$p \wedge q \therefore p$	It is sunny and warm. $\therefore$ It is sunny.
Conjunction	$p, q \therefore p \wedge q$	It is sunny. It is warm. $\therefore$ Sunny and warm.
Resolution	$p \vee q, \neg p \vee r \therefore q \vee r$	Core of automated theorem proving.
QUANTIFIED STATEMENT RULES		
Universal Instantiation	$\forall x P(x) \therefore P(c)$	All humans are mortal. $\therefore$ Socrates is mortal.
Universal Generalization	$P(c) \text{ for any } c \therefore \forall x P(x)$	If shown true for arbitrary c, true for all.
Existential Instantiation	$\exists x P(x) \therefore P(c) \text{ for some } c$	Someone won the lottery. Call them c.
Existential Generalization	$P(c) \therefore \exists x P(x)$	5 is prime. $\therefore$ There exists a prime number.

Use in Building Arguments

What Is an Argument?

In logic, an **argument** is a sequence of statements (called **premises**) that are claimed to support another statement (called the **conclusion**).

Structure of an argument:

Premise1, Premise2, Premise3, ..., Conclusion

Example:

- Premise 1: If it rains, the ground gets wet.
- Premise 2: It is raining.
- Conclusion: Therefore, the ground gets wet.

Fallacies

A **fallacy** is an error in reasoning that makes an argument invalid. A fallacious argument may have true premises and even a true conclusion, but the **logical connection** between premises and conclusion is flawed.

Key distinction:

- A **valid argument** guarantees: If premises are true, conclusion **must** be true.
- A **fallacious argument** does **not** guarantee this. The conclusion could be false even if all premises are true.

Important: An argument can have a **true conclusion** but still be **fallacious** (invalid). Truth of conclusion does not rescue a bad argument.

Example of fallacious argument with true conclusion:

- Premise 1: All dogs are mammals.
- Premise 2: Cats are mammals.
- Fallacious conclusion: Therefore, all dogs are cats. (True? No, false.)

Better example of fallacious but accidentally true conclusion:

- Premise 1: If it rains, the ground gets wet.
- Premise 2: The ground is wet.
- Fallacy: Therefore, it rained. (Conclusion might be true, but the reasoning is invalid — sprinklers could have wet the ground.)

Formal Fallacies

These are invalid argument forms that **look like** valid rules of inference but are not.

Fallacy 1: Affirming the Consequent

Invalid Form	Looks like	Why It's Wrong
$p \rightarrow q$	Modus Ponens (valid) is $p \rightarrow q, p \therefore q$	q could be true even if p is false. The implication only tells you what happens when p is true, not when q is true.
q	Affirming the consequent swaps premise and conclusion incorrectly.	
$\therefore p$		

Example:

- Premise 1: If it is raining, then the ground is wet. ($p \rightarrow q$)
- Premise 2: The ground is wet. (q)
- Fallacious Conclusion: Therefore, it is raining. (p)

Why it's fallacious: The ground could be wet from sprinklers, a hose, a broken pipe, etc. The conclusion does **not** necessarily follow.

Truth table demonstration:

p	q	$p \rightarrow q$	When premises are true?	Conclusion p ?
T	T	T	✓	T (works in this row)
F	T	T	✓	F (premises true, conclusion false!)

The second row shows the problem: p is false, q is true, $p \rightarrow q$ is true. Both premises true, conclusion false $\rightarrow$ invalid.

Fallacy 2: Denying the Antecedent

Invalid Form	Looks like	Why It's Wrong
$p \rightarrow q$	Modus Tollens (valid) is $p \rightarrow q, \neg q \therefore \neg p$	q could still be true even if p is false. The implication says nothing about what happens when p is false.
$\neg p$	Denying the antecedent incorrectly concludes the negation of q .	

$\therefore \neg q$

Example:

- Premise 1: If it is raining, then the ground is wet. ($p \rightarrow q$)
- Premise 2: It is not raining. ($\neg p$)
- Fallacious Conclusion: Therefore, the ground is not wet. ($\neg q$)

Why it's fallacious: The ground could be wet from other sources. The absence of rain does not guarantee dry ground.

Truth table demonstration:

p	q	$p \rightarrow q$	$\neg p$	When premises true?	Conclusion $\neg q$?
F	T	T	T	✓	F (premises true, conclusion false!)

Again, the premises can be true while the conclusion is false.

Fallacy 3: The Fallacy of the Converse

This is essentially the same as Affirming the Consequent, but with quantifiers.

Invalid Form Correct Form

$\forall x(P(x) \rightarrow Q(x)) \quad \forall x(P(x) \rightarrow Q(x))$

$Q(a) \quad P(a)$

Invalid Form Correct Form

$\therefore P(a)$ (Invalid) $\therefore Q(a)$ (Valid — Modus Ponens / UI)

Example:

- Premise 1: All dogs are mammals. ($\forall x(Dog(x) \rightarrow Mammal(x))$)
- Premise 2: Fluffy is a mammal. ($Mammal(Fluffy)$)
- Fallacious Conclusion: Therefore, Fluffy is a dog. ($Dog(Fluffy)$)

Why it's fallacious: Fluffy could be a cat, a horse, a human, etc.

Fallacy 4: The Fallacy of the Inverse

Same as Denying the Antecedent, with quantifiers.

Invalid Form Correct Form

$\forall x(P(x) \rightarrow Q(x))$ $\forall x(P(x) \rightarrow Q(x))$

$\neg P(a)$ $\neg Q(a)$

$\therefore \neg Q(a)$ (Invalid) $\therefore \neg P(a)$ (Valid — Modus Tollens / UI)

Example:

- Premise 1: All dogs are mammals.
- Premise 2: Fluffy is not a dog.
- Fallacious Conclusion: Therefore, Fluffy is not a mammal.

Why it's fallacious: Fluffy could be a cat (which is a mammal).

Fallacy 5: Quantifier Swap Fallacy

Invalid Form

Why It's Wrong

$\forall x \exists y P(x, y)$

Order matters! $\forall x \exists y$ means "for each x, there is some y (possibly different for each x)."

$\therefore \exists y \forall x P(x, y)$

$\exists y \forall x$ means "there is a single y that works for all x" — a much stronger statement.

Example:

- Premise: Every person has a mother. ($\forall x \exists y (MotherOf(y,x))$)
- Fallacious Conclusion: There is a person who is the mother of everyone. ($\exists y \forall x (MotherOf(y,x))$)

Why it's fallacious: The premise is true (everyone has a mother), but the conclusion is false (no single person is the mother of all people).

Fallacy 6: Existential Instantiation Violation

Using the same constant for two different existential instantiations incorrectly.

Invalid reasoning:

- Premise 1: $\exists x P(x)$ (Some student is late.)
- Premise 2: $\exists x Q(x)$ (Some student is absent.)
- Invalid step: Let c be such that $P(c)$ (from premise 1).
- Invalid step: Let c be such that $Q(c)$ (from premise 2 — using same c !).
- Fallacious conclusion: $\exists x (P(x) \wedge Q(x))$ (Some student is both late and absent.)

Why it's fallacious: The student who is late might not be the same student who is absent. You must use **different** constants for different existential instantiations unless you have proven they refer to the same individual.

Correct approach: Use c for the first, d for the second. Without additional premises linking c and d , you cannot conclude $P(c) \wedge Q(c)$.

Fallacy 7: Universal Generalization on a Non-Arbitrary Constant

Invalid Form

Why It's Wrong

$P(c)$ where c is a specific individual

To generalize to $\forall x P(x)$, c must be **arbitrary** (no special assumptions). A specific named individual (e.g., "Socrates") is not arbitrary.

$\therefore \forall x P(x)$

Example:

- Premise: Socrates is mortal. ($Mortal(Socrates)$)
- Invalid conclusion: Therefore, all humans are mortal. ($\forall x (Human(x) \rightarrow Mortal(x))$)

Informal Fallacies

While formal fallacies violate logical structure, informal fallacies violate rules of relevance, clarity, or evidence.

1. *Ad Hominem (Attack on the Person)*

Attacking the character, motive, or background of the person making an argument instead of addressing the argument itself.

Example:

- "You say that we should reduce carbon emissions, but you drive a car. Therefore, your argument is wrong."

Why it's fallacious: The truth of an argument does not depend on the personal behavior of the speaker.

2. *Straw Man*

Misrepresenting someone's argument to make it easier to attack.

Example:

- Person A: "We should spend more on public transportation."
- Person B: "So you want to eliminate all cars and ban private vehicles? That's ridiculous."

Why it's fallacious: Person B is attacking a distorted version of the argument, not the actual argument.

3. *Appeal to Authority (Argumentum ad Verecundiam)*

Claiming something is true because an authority figure says it, when the authority is not an expert in that domain.

Example:

- "My favorite actor says this brand of toothpaste is the best, so it must be true."

Why it's fallacious: An actor is not an expert in dentistry.

Note: Using a legitimate expert in their field is **not** a fallacy (e.g., citing a cardiologist about heart health).

4. *Appeal to Ignorance (Argumentum ad Ignorantiam)*

Arguing that a claim is true because it has not been proven false, or false because it has not been proven true.

Example:

- "No one has proven that aliens don't exist, so aliens exist."

Why it's fallacious: Lack of evidence against a claim is not evidence for the claim.

5. *Circular Reasoning (Petitio Principii)*

Assuming what you are trying to prove as a premise.

Example:

- "The Bible is true because it is the word of God. We know it is the word of God because the Bible says so."

Why it's fallacious: The conclusion is used as a premise.

6. False Dilemma (False Dichotomy)

Presenting only two options when more exist.

Example:

- "You're either with us or against us."

Why it's fallacious: There may be neutral positions, partial agreement, etc.

7. Slippery Slope

Arguing that a small first step will inevitably lead to an extreme (and undesirable) outcome without evidence.

Example:

- "If we allow students to use calculators, soon they will forget how to do any basic math, and eventually our entire education system will collapse."

Why it's fallacious: No evidence that the chain of events will actually occur.

8. Hasty Generalization

Drawing a broad conclusion from a small or unrepresentative sample.

Example:

- "I met two people from that city, and they were both rude. Therefore, everyone from that city is rude."

Why it's fallacious: Sample size is too small to support the conclusion.

9. Post Hoc Ergo Propter Hoc ("After this, therefore because of this")

Assuming that because event A happened before event B, A caused B.

Example:

- "I wore my lucky socks, and my team won. Therefore, my lucky socks caused the win."

Why it's fallacious: Correlation (or temporal sequence) does not imply causation.

10. Begging the Question (another name for Circular Reasoning)

See Circular Reasoning above.

Common Formal Fallacies

Fallacy Name	Invalid Form	Valid Form (for comparison)
Affirming the Consequent	$p \rightarrow q, q \therefore p$	$p \rightarrow q, p \therefore q$ (MP)
Denying the Antecedent	$p \rightarrow q, \neg p \therefore \neg q$	$p \rightarrow q, \neg q \therefore \neg p$ (MT)
Fallacy of the Converse	$\forall x(P(x) \rightarrow Q(x)), Q(a) \therefore P(a)$	$\forall x(P(x) \rightarrow Q(x)), P(a) \therefore Q(a)$
Fallacy of the Inverse	$\forall x(P(x) \rightarrow Q(x)), \neg P(a) \therefore \neg Q(a)$	$\forall x(P(x) \rightarrow Q(x)), \neg Q(a) \therefore \neg P(a)$
Quantifier Swap Fallacy	$\forall x \exists y P(x, y) \therefore \exists y \forall x P(x, y)$	Not equivalent; order matters
EI Constant Reuse	Using same constant for two different $\exists$ instantiations	Use fresh constants
UG on Non-Arbitrary	$P(c)$ (specific) $\therefore \forall x P(x)$	$P(c)$ (arbitrary) $\therefore \forall x P(x)$

Basic Terminologies

Why These Terminologies Matter?

Before you can prove anything, you need to understand the language of proof. These terms describe the **status** of statements (Is it an axiom? A theorem? A conjecture?), the **relationships** between statements (What is a lemma? A corollary?), and the **building blocks** of logical systems (What is an axiom? A rule of inference?).

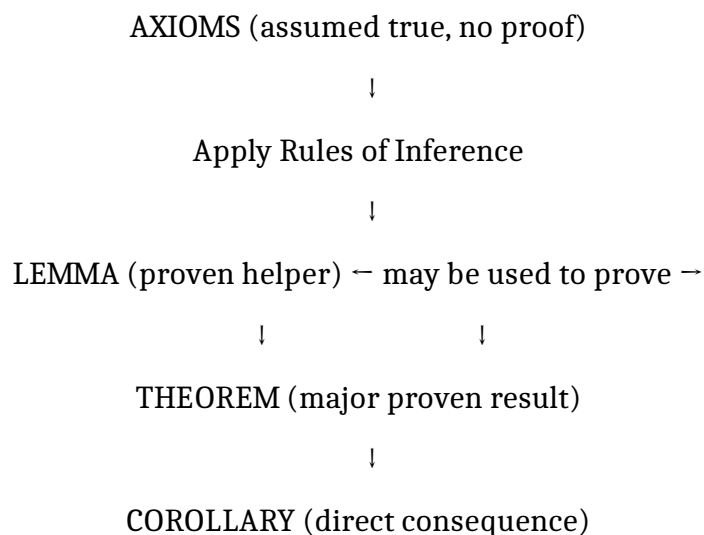
Think of this as learning the "job titles" for statements. Just as you need to know what a "manager," "director," or "intern" means in an organization, you need to know what "theorem," "lemma," "corollary," and "conjecture" mean in mathematics.

The Hierarchy of Mathematical Statements

From least certain to most certain:

Term	Certainty	Definition	Example
Conjecture	Low (unproven)	A statement believed to be true but not yet proven.	"Every even number greater than 2 is the sum of two primes." (Goldbach's conjecture)
Lemma	High (proven)	A helper theorem used to prove a larger, more important theorem.	"If n is even, then n^2 is even." (Used to prove something about odd numbers)
Theorem	High (proven)	A major, important statement that has been proven true.	"The square root of 2 is irrational."
Corollary	High (proven)	A statement that follows directly from a theorem, often a special case.	"The square root of 8 is irrational." (Follows from $\sqrt{2}$ irrational)
Axiom (Postulate)	Foundational (assumed)	A statement accepted as true without proof. The starting point of a logical system.	"Things equal to the same thing are equal to each other." (Euclid's axiom)

Visual hierarchy:



1. Axiom (Postulate)

Definition: An axiom (or postulate) is a statement that is taken to be true without proof. Axioms serve as the foundational assumptions of a logical system. Every theorem in that system is proven from these axioms using rules of inference.

Characteristics:

- Self-evident?** Not necessarily. Axioms are chosen because they are useful and consistent, not necessarily "obvious."
- Undecidable within the system:** You cannot prove an axiom from other axioms (otherwise it wouldn't be an axiom).
- Different systems have different axioms:** Euclidean geometry has different axioms than non-Euclidean geometry.

Examples:

- Logic:** "A proposition is either true or false (not both)." (Law of Excluded Middle)
- Euclidean Geometry:** "A straight line can be drawn between any two points."
- Arithmetic (Peano Axioms):** "0 is a natural number." "Every natural number has a successor."

In this course: You will assume standard logical axioms (tautologies) and rules of inference without proving them.

2. Theorem

Definition: A theorem is an important statement that has been proven to be true using axioms, definitions, and previously established theorems.

Characteristics:

- Proven:** A theorem is not a guess or conjecture. It has a rigorous proof.
- Significant:** The term "theorem" is usually reserved for major, interesting, or non-trivial results.
- Published:** Theorems are the main results of mathematical research.

Examples:

- Pythagorean Theorem:** In a right triangle, $a^2 + b^2 = c^2$.
- Fundamental Theorem of Arithmetic:** Every integer greater than 1 is either prime or a unique product of primes.
- Pythagorean Theorem (in logic):** $\neg \forall x P(x) \equiv \exists x \neg P(x)$ (De Morgan's law for quantifiers — can be proven).

Structure of a theorem statement:

- Usually stated as an implication: "If [hypothesis], then [conclusion]."

•Or a universal statement: "For all [things], [property holds]."

Example: "If n is an odd integer, then n^2 is odd."

3. Lemma

Definition: A lemma is a helper theorem — a proven statement that is not necessarily important on its own but is used as a stepping stone to prove a larger, more important theorem.

Characteristics:

- Subsidiary:** Lemmas serve the main theorem.
- Often technical:** Lemmas may be small, specific, and technical.
- Inside proofs:** You will often see "Lemma 1" followed by "Proof of Main Theorem."

Examples:

- Lemma (for proving $\sqrt{2}$ is irrational):** "If n^2 is even, then n is even."
- Lemma (for proving the Euclidean algorithm works):** "If $a=bq+r$, then $\gcd(a,b)=\gcd(b,r)$."

Why call it a lemma instead of a theorem? To signal that it is a tool, not the final destination. If a lemma turns out to be very important, it may later be called a theorem.

4. Corollary

Definition: A corollary is a statement that follows **directly and immediately** from a theorem, often a special case or a simple consequence.

Characteristics:

- Short proof:** The proof is usually a few lines, referencing the main theorem.
- Often stated immediately after the theorem.**
- Sometimes obvious:** A corollary may be "obvious" once the theorem is understood.

Examples:

- Theorem:** "The sum of the angles in any triangle is 180° ."
- Corollary:** "No triangle can have two right angles." (If it did, the sum would exceed 180° .)
- Theorem:** "If n is odd, then n^2 is odd."
- Corollary:** "If n is odd, then n^4 is odd." (Since $n^4=(n^2)^2$, and n^2 is odd by the theorem, apply the theorem again.)

5. Conjecture

Definition: A conjecture is a statement that is believed to be true based on evidence (examples, patterns, intuition) but has **not yet been proven**.

Characteristics:

- Unproven:** No rigorous proof exists (or the proof has not been verified).
- May be true or false:** Some conjectures turn out to be true (and later become theorems). Some turn out to be false (disproven by a counterexample).
- Drives research:** Mathematicians work to prove or disprove conjectures.

Famous Conjectures:

Conjecture	Statement	Status
Goldbach's Conjecture	Every even number greater than 2 is the sum of two primes.	Unproven (open since 1742)
Riemann Hypothesis	The nontrivial zeros of the Riemann zeta function have real part $1/2$.	Unproven (Millennium Problem)
Twin Prime Conjecture	There are infinitely many pairs of primes that differ by 2.	Unproven
P vs NP	Can every problem whose solution can be verified quickly also be solved quickly?	Unproven (Millennium Problem)

Relationships Between Terminologies

From	To	Relationship
Axioms	Theorems	Axioms are assumed; theorems are proven from axioms.
Lemma	Theorem	Lemmas are used to prove theorems.
Theorem	Corollary	Corollaries are immediate consequences of theorems.

From	To	Relationship
Conjecture	Theorem	A conjecture becomes a theorem when proven true.
Conjecture	Disproof	A conjecture is disproven by a single counterexample.
Proof	Theorem	A proof establishes a theorem.
Tautology	Logical truth	Always true by logical form alone.
Contradiction	Logical falsehood	Always false by logical form alone.

Key takeaways:

- **Axioms** are the starting points (assumed true).
- **Theorems** are major results (proven true).
- **Lemmas** are helper theorems (proven true, but subsidiary).
- **Corollaries** are direct consequences (proven true, follow from theorems).
- **Conjectures** are statements awaiting proof (not yet known to be true).

Different Proof Techniques

A **proof** is a logical argument that establishes the truth of a statement (theorem) from axioms, definitions, and previously established theorems using valid rules of inference.

Characteristics of a good proof:

- **Valid:** Every step follows logically from previous steps.
- **Complete:** No gaps in reasoning.
- **Clear:** Easy to follow and understand.
- **Rigorous:** No hidden assumptions.

Overview of Proof Techniques

Proof Technique	Basic Idea	When to Use
Direct Proof	Assume hypothesis, derive conclusion step by step.	Statements of the form $p \rightarrow q$ where you can connect p to q directly.

Proof Technique	Basic Idea	When to Use
Indirect Proof (Proof by Contraposition)	Prove $\neg q \rightarrow \neg p$ instead of $p \rightarrow q$.	When the contrapositive is easier to prove.
Proof by Contradiction	Assume the negation of the conclusion, derive a contradiction.	When assuming the opposite leads to an obvious impossibility.
Proof by Mathematical Induction	Prove base case + inductive step to show $\forall n P(n)$.	Statements about natural numbers (loops, recursive algorithms, sums, inequalities).
Proof by Cases	Break the statement into exhaustive cases, prove each separately.	When different situations require different reasoning.
Constructive Proof	Provide an explicit example or construction.	Proving existence statements ($\exists x P(x)$).
Non-Constructive Proof	Show existence without providing an explicit example.	When you can prove something must exist without finding it.
Proof by Exhaustion	Check a finite number of possibilities.	When the domain is finite and small enough to enumerate.

Direct Proof

Definition: A direct proof assumes the hypothesis (antecedent) and uses logical steps, definitions, and known theorems to derive the conclusion.

When to use: When you can directly connect the hypothesis to the conclusion through a chain of logical implications.

Example 1: Prove: If n is an even integer, then n^2 is even.

Proof:

1. Assume n is even. (Hypothesis)
2. By definition of even, $n=2k$ for some integer k .
3. Then $n^2=(2k)^2=4k^2=2(2k^2)$.

4. Since $2k^2$ is an integer, n^2 is even (by definition of even).

5. Therefore, if n is even, then n^2 is even. ✓

Example 2: Prove: If x is an odd integer, then x^2 is odd.

Proof:

1. Assume x is odd. (Hypothesis)

2. By definition of odd, $x=2k+1$ for some integer k .

3. Then $x^2=(2k+1)^2=4k^2+4k+1=2(2k^2+2k)+1$.

4. Since $2k^2+2k$ is an integer, x^2 is odd (by definition of odd).

5. Therefore, if x is odd, then x^2 is odd. ✓

Example 3 (Computer Science): Prove: If a binary number ends in 0, it is even.

Proof:

1. Let N be a binary number ending in 0.

2. In binary, the least significant bit (LSB) represents $2^0=1$.

3. If the LSB is 0, then N is a multiple of 2 (no contribution of 1).

4. By definition, a number divisible by 2 is even.

5. Therefore, N is even. ✓

Indirect Proof (Proof by Contraposition)

Definition: Instead of proving $p \rightarrow q$, we prove its **contrapositive** $\neg q \rightarrow \neg p$, which is logically equivalent.

Why use it? Sometimes $\neg q \rightarrow \neg p$ is easier to prove directly than $p \rightarrow q$.

Example 1: Prove: If n^2 is odd, then n is odd.

Direct proof attempt: Hard to go from n^2 being odd directly to n being odd.

Proof by contraposition:

1. The contrapositive of "If n^2 is odd, then n is odd" is "If n is not odd, then n^2 is not odd."

2. "Not odd" means "even". So we prove: If n is even, then n^2 is even.

3. Assume n is even. Then $n=2k$ for some integer k .

4. Then $n^2=(2k)^2=4k^2=2(2k^2)$, which is even.

5. Therefore, the contrapositive is true, so the original statement is true. ✓

Example 2: Prove: If $x+y \geq 10$, then $x \geq 5$ or $y \geq 5$.

Proof by contraposition:

1. Contrapositive: If $\text{NOT}(x \geq 5 \text{ or } y \geq 5)$, then $\text{NOT}(x+y \geq 10)$.

2. By De Morgan: $\neg(x \geq 5 \vee y \geq 5) \equiv (x < 5) \wedge (y < 5)$.

3. So we prove: If $x < 5$ and $y < 5$, then $x+y < 10$.

4. Assume $x < 5$ and $y < 5$.

5. Then $x+y < 5+5=10$.

6. Therefore, the contrapositive is true, so the original statement is true. ✓

Example 3: Prove: If n^2 is even, then n is even.

Proof by contraposition:

1. Contrapositive: If n is odd, then n^2 is odd.

2. Assume n is odd: $n=2k+1$.

3. Then $n^2=(2k+1)^2=4k^2+4k+1=2(2k^2+2k)+1$, which is odd.

4. Therefore, the original statement is true. ✓

Proof by Contradiction (Reductio ad Absurdum)

Definition: To prove a statement P , we assume that P is **false** (i.e., assume $\neg P$) and derive a logical contradiction (a statement that is always false, like $Q \wedge \neg Q$). Since the assumption leads to contradiction, the assumption must be false, so P is true.

When to use: When direct proof is difficult, or when proving that something **cannot** happen (impossibility proofs), or when proving that something **exists** (by contradiction that nothing exists).

Example 1 (Classic): Prove that $\sqrt{2}$ is irrational.

Proof by contradiction:

1. Assume, for contradiction, that $\sqrt{2}$ is rational.

2. Then $\sqrt{2} = \frac{a}{b}$ where a and b are integers with no common factors (fraction in lowest terms), and $b \neq 0$.

3. Square both sides: $2 = \frac{a^2}{b^2} \rightarrow a^2 = 2b^2$.

4. This means a^2 is even, so a is even (by earlier lemma: if a square is even, the number is even).

5. Let $a=2k$ for some integer k .

6. Substitute: $(2k)^2 = 2b^2 \rightarrow 4k^2 = 2b^2 \rightarrow 2k^2 = b^2$.

7. This means b^2 is even, so b is even.

8. Then both a and b are even, so they have a common factor of 2.

9. This contradicts the assumption that a and b have no common factors.

10. Therefore, our assumption is false. Hence, $\sqrt{2}$ is irrational. ✓

Example 2: Prove that there are infinitely many prime numbers.

Proof by contradiction (Euclid's proof):

1. Assume, for contradiction, that there are only finitely many primes: $p_1, p_2, \dots, p_n$.

2. Consider the number $N = p_1 \times p_2 \times \dots \times p_n + 1$.

3. When divided by any prime p_i , N leaves a remainder of 1 (since $N - 1$ is divisible by p_i).

4. So N is not divisible by any existing prime.

5. Therefore, N must either be prime itself (a new prime not in the list) or have a prime factor not in the list.

6. In either case, we have found a prime not in the original finite list.

7. This contradicts the assumption that the list contained all primes.

8. Therefore, there are infinitely many primes. ✓

Example 3: Prove that if $3x+2$ is odd, then x is odd.

Proof by contradiction:

1. Assume, for contradiction, that $3x+2$ is odd AND x is even (negation of conclusion: x is not odd, so x is even).

2. If x is even, then $x = 2k$ for some integer k .

3. Then $3x+2 = 3(2k)+2 = 6k+2 = 2(3k+1)$, which is even.

4. This contradicts the premise that $3x+2$ is odd.

5. Therefore, our assumption (that x is even) must be false. Hence, x is odd. ✓

Proof by Mathematical Induction

Definition: Induction is a proof technique for proving statements of the form $\forall n P(n)$ where n ranges over natural numbers (or a well-ordered set).

Principle of Mathematical Induction:

If we can show:

1. **Base Case:** $P(1)$ is true (or $P(0)$, depending on the starting point).

2. Inductive Step: For an arbitrary $k \geq 1$, if $P(k)$ is true (the **inductive hypothesis**), then $P(k+1)$ is true.

Then $P(n)$ is true for **all** natural numbers n .

Variants:

- **Strong Induction:** Assume $P(1), P(2), \dots, P(k)$ are all true, then prove $P(k+1)$.
- **Complete Induction:** Same as strong induction.
- **Second Principle of Induction:** Used for statements where $P(k+1)$ depends on multiple previous cases.

Example 1 (Sum of natural numbers): Prove that for all $n \geq 1$,
 $1+2+3+\dots+n=2n(n+1)$

Proof by induction:

- **Base Case ($n = 1$):** Left side = 1. Right side = $2 \cdot 1(1+1) = 2 \cdot 1 = 2$. ✓
- **Inductive Step:** Assume the formula holds for $n=k$ (inductive hypothesis):
 $1+2+\dots+k=2k(k+1)$
Now prove for $n=k+1$:
 $1+2+\dots+k+(k+1)=2k(k+1)+(k+1)$
 $= (k+1)(2k+1) = (k+1)(2k+2) = 2(k+1)(k+1)$
Which is exactly the formula for $n=k+1$.
- **Conclusion:** By induction, the formula holds for all $n \geq 1$. ✓

Example 2 (Sum of squares): Prove for all $n \geq 1$:
 $1^2+2^2+\dots+n^2=6n(n+1)(2n+1)$

Proof by induction:

- **Base Case ($n = 1$):** Left = 1. Right = $6 \cdot 1 \cdot 2 \cdot 3 = 6 \cdot 1 = 6$. ✓
- **Inductive Step:** Assume true for $n=k$:
 $1^2+\dots+k^2=6k(k+1)(2k+1)$
Prove for $n=k+1$:
 $1^2+\dots+k^2+(k+1)^2=6k(k+1)(2k+1)+(k+1)^2$
 $= (k+1)[6k(2k+1)+(k+1)] = (k+1)[6 \cdot 2k^2+k+6k+6]$
 $= (k+1)[6 \cdot 2k^2+7k+6] = (k+1)[6(k+2)(2k+3)] = 6(k+1)(k+2)(2k+3)$
Which is the formula for $n=k+1$. ✓
- **Conclusion:** Formula holds for all $n \geq 1$.

Example 3 (Computer Science - Loop correctness): Prove that the following algorithm correctly computes the sum of first n integers.

Proof by induction on number of iterations:

- **Base Case:** After 0 iterations, $s=0$ and $i=1$. The invariant $s=1+2+\dots+(i-1)$ holds (empty sum = 0).
- **Inductive Step:** Assume before iteration $i=k$, $s=1+\dots+(k-1)$. After executing $s = s + k$ and $i = k + 1$, $s=1+\dots+k$ and $i=k+1$. Invariant holds.
- **Conclusion:** When loop exits ($i=n+1$), $s=1+\dots+n$. The algorithm is correct.

Proof by Cases:

Definition: When a statement can be broken into a finite number of cases that cover all possibilities, we prove each case separately.

Example 1 Prove that for any integer n , n^2+n is even.

Proof by cases:

- **Case 1:** n is even.
- Then $n=2k$ for some integer k .
- $n^2+n=(4k^2)+2k=2(2k^2+k)$, which is even.
- **Case 2:** n is odd.
- Then $n=2k+1$ for some integer k .
- $n^2+n=(4k^2+4k+1)+(2k+1)=4k^2+6k+2=2(2k^2+3k+1)$, which is even.
- Since every integer is either even or odd (exhaustive cases), n^2+n is even for all integers n . ✓

Example 2: Prove that $|x \cdot y| = |x| \cdot |y|$ for all real numbers x, y .

Proof by cases (based on signs of x and y):

- **Case 1:** $x \geq 0, y \geq 0 \rightarrow |xy| = xy = |x| |y|$
- **Case 2:** $x \geq 0, y < 0 \rightarrow |xy| = |x \cdot (\text{negative})| = -xy = x(-y) = |x| |y|$
- **Case 3:** $x < 0, y \geq 0 \rightarrow |xy| = |(\text{negative}) \cdot y| = -xy = (-x)y = |x| |y|$
- **Case 4:** $x < 0, y < 0 \rightarrow |xy| = |(\text{positive})| = xy = (-x)(-y) = |x| |y|$
- All cases covered, so statement holds. ✓

Example 3: Prove that the square of any integer leaves remainder 0 or 1 when divided by 4.

Proof by cases:

- **Case 1:** n is even. $n=2k \rightarrow n^2=4k^2$, remainder 0.
- **Case 2:** n is odd. $n=2k+1 \rightarrow n^2=4k^2+4k+1=4(k^2+k)+1$, remainder 1.
- Since every integer is even or odd, the statement holds. ✓

Direct proof

Assume p is true, then use logical steps to show q must be true.

Goal: $p \rightarrow q$. Assume p . Derive q step by step.

Example: Prove "if n is even, n^2 is even." Assume $n = 2k$, then $n^2 = 4k^2 = 2(2k^2)$, which is even. ✓

Indirect / Contrapositive Proof by contraposition

Since $p \rightarrow q \equiv \neg q \rightarrow \neg p$, prove the contrapositive instead — often easier.

Goal: $p \rightarrow q$. Instead prove $\neg q \rightarrow \neg p$.

Example: Prove "if n^2 is odd, then n is odd." Contrapositive: if n is even, n^2 is even — which is the direct proof above. ✓

Contradiction Proof by contradiction (Reductio ad absurdum)

Assume the negation of what you want to prove, then derive a contradiction (something that is false). The negated assumption must therefore be false.

Goal: prove p . Assume $\neg p$. Derive a contradiction ($\perp$). $\therefore p$ must be true.

Example: Prove $\sqrt{2}$ is irrational. Assume it equals a/b in lowest terms. Then $2b^2 = a^2$, so a is even, meaning b is also even — contradicting that a/b is in lowest terms. ✓

Induction Proof by mathematical induction

Used to prove statements about all positive integers. Works like an infinite chain of dominoes.

Step 1 (Base case): Prove $P(1)$ is true. **Step 2 (Inductive step):** Assume $P(k)$ is true (inductive hypothesis). Prove $P(k+1)$ is also true. **Conclusion:** $P(n)$ is true for all $n \geq 1$.

Example: Prove $1+2+3+\dots+n = n(n+1)/2$. Base: $n=1 \rightarrow 1 = 1(2)/2 = 1$ ✓. Inductive step: Assume true for k . Then sum to $k+1 = k(k+1)/2 + (k+1) = (k+1)(k+2)/2$. ✓